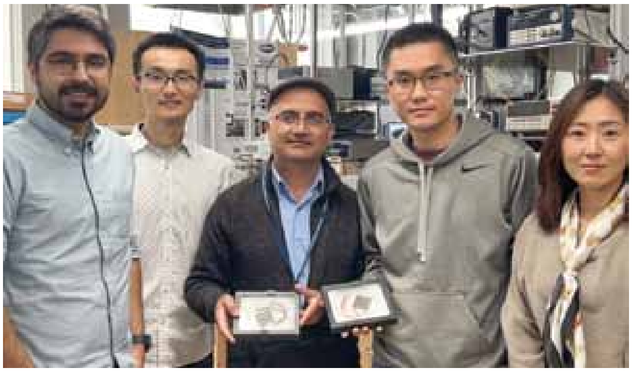
by harmonic distortions, can amplify the source signal. The Fourier transform detects piezoelectric testing interferences caused by friction and identifies phantom energy using a simple technique. Energy spikes can cause overload, and signal filters can eliminate interference. The technique can help manufacturers create precise devices for various applications and extend their lifetimes.

A unique materials design can boost conversion efficiency of thermoelectric devices

Scientists at Penn State and the National Renewable Energy Laboratory have developed a unique materials design that can push the conversion efficiency of thermoelectric devices up to 15%. Researchers created functionally graded thermoelectric materials with 15.2% efficiency in a single-leg device, surpassing commercially available devices. This



Amin Nozariasbmarz, Yu Zhang, Bed Poudel, Wenjie Li and Na Liu, left to right, researchers in the Department of Materials Science and Engineering at Penn State, display the thermoelectric modules they created (Credit: Pennsylvania State University)

approach eliminates non-uniform interfaces, improving efficiency and enabling greener technologies to address global greenhouse gas emissions. New method combines materials for high and low temp sides, with no need for an interface. Electrical and mechanical field-assisted sintering used to produce materials by layering powders and adding dopants. Similar materials from the same family with optimised chemical compositions used to sinter at the same temperature in one step.

A new soft robot platform helps create customised machines for specific tasks

Researchers at MIT have developed a soft robot co-design platform that can test optimal shapes and sizes for robotic performance in different environments. SoftZoo's versatility reduces simulation costs for computational control and design problems, optimising co-design, leading to more sophisticated algorithms for designing and moving soft robots. The system's terrain interaction simulation highlights the significance of morphology, where optimal biological structures vary per environment. SoftZoo co-optimises land and aquatic machines, enhancing their behavioural and morphological intelligence, valuable in rescue missions and exploration. Its open source simulation accelerates the movement of flexible robots in different environments, aiding designers to create customised machines for specific tasks, such as efficiently searching floods.

Low-cost smart textiles with LEDs, sensors, and energy harvesting developed

Researchers at the University of Cambridge have developed low-cost smart textiles with LEDs, sensors, and energy harvesting that can be made in any shape or size using regular clothing machines. Smart textile manufacturing requires a large investment and is limited to 30cm diameter

Researchers have developed next-generation smart textiles—incorporating LEDs, sensors, energy harvesting, and storage—that can be produced inexpensively, in any shape or size, using the same machines used to make the clothing we wear every day (Credit: Sanghyo Lee)

due to use of integrated circuit wafers. To overcome this, researchers coated textile fibres, making them stretchable and weavable, resulting in a 117cm display. Now, automated processes can create smart textiles of any size and shape, with energy storage, LEDs, and transistors woven into fibres and interconnected via laser welding. Optimised processes make durable smart textiles that withstand industrial weaving machines. Encapsulation methods preserve functionality, while mechanical force and thermal energy automate weaving and interconnection. Researchers partnered with textile manufacturers and produced test patches, scalable to larger sizes for mass production.